

Fifteenth century CE Bolivian maize reveals genetic affinities with ancient Peruvian maize

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eLife Assessment

This **useful** study attempts to place an ancient maize sample from Bolivia, dated to the end of the Incan empire, in genetic and geographical context. The analyses show that this sample is most closely related to ancient Peruvian maize, but the data remain **inadequate** to determine the direction of dispersal and the extent of Inca influence over the genetic make up of the analyzed sample. There are additional deficiencies in the statistical analyses and selection inferences. The topic of the study would appeal to researchers studying maize dispersal and adaptation.

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Abstract

Previous archaeological and anthropological studies have demonstrated the myriad of ways that cultural and political systems shape access to food and food preferences. However, few studies have conducted a biocultural analysis linking specific genotypic/phenotypic traits as evidence of cultural selection in ancient contexts. Here, we provide insight into this topic through ancient genome data from Bolivian maize dating to ~500-600 BP, included as an offering with the mummified remains of a young girl. These data are compared to 16 previously published archaeological maize samples spanning at least 5,000 years of evolution, and 226 modern maize samples. Our phylogenetic analysis showed that the archaeological Bolivian maize (aBM) has the closest genetic distance to the archaeological maize from ancient Peru, which in turn shared the most similarities with archaeological Peruvian maize. During the period of interaction between the Inca state and local polities in the central Andes and consequent interactions with local agricultural traditions, the genetic diversity of maize increased. Ovule development in modern maize was selected and compared to those in archaeological specimens, revealing evidence of targeted breeding strategies aimed at improving seed quality and yield. While the cultural origin of the maize –

either Inca or local Aymara – is uncertain, we demonstrate that the samples are most similar to Peruvian maize and potential targeted selection strategies for enhanced growth were well established by the 15th century.

Significance Statement

Maize has been improved through selective breeding delivering desirable horticultural, agronomic, and cultural characteristics, related to environment, nutrition, etc. Inca culture was closely associated with maize diversity, but there is little evidence to show how and why culture influenced that diversity and modern maize biogeographic patterns in South America. We show that in the 15th century CE, maize diversity in the central Andes was enhanced through selection for specific traits associated with reproduction. We hypothesize that this may have contributed to maize becoming a staple under the Inca, as happened previously with expanding Tiwanaku interactions and influence during the Middle Horizon.

Introduction

Modern maize (*Zea mays* ssp. *mays* L.) is one of the most important and productive global food crops¹. Prior to European contact in the Americas, maize had been adapted for a wide range of environments. Its versatility is clear from its extensive geographic distribution today, ranging from 58° north latitude to 40° south latitude, and growing below sea level as well as 3,600 meters above sea level in the Andes². In Bolivia, there is empirical evidence that small maize varieties can grow at altitudes of 3,800–3,900 meters above sea level (masl) in some areas of the Lake Titicaca Basin³, as well as in other sheltered areas of Oruro and Potosí, although with lower levels of production^{4, 5}. Studies of archaeological samples and modern molecular data indicate that maize originally evolved in Mexico from the flowering species of grass, teosinte (*Zea mays* ssp. *parviglumus*) ~9,000 years before the present (cal. BP)^{6, 7} and spread into South America by ~7,000 cal BP^{8, 9} from the lowland and Eastern North America by ~2500 cal BP¹⁰. During this domestication process there was hybridization between maize and mexicana (*Zea mays* ssp. *mexicana*)¹¹. The dispersal of maize to the rest of the world followed European colonization of the Americas in the 15th and 16th centuries AD^{12, 13}. The evolution and diversity of maize has been shaped by both cultural and natural selection in a wide range of environmental and social contexts. For example, previous research has shown that temperate adaption in modern maize is characterized by early flowering¹³, evidence has also shown that cultural preferences were a factor in maintaining otherwise undesirable traits^{14, 15}. Maize seeds were selected and retained by farmers for use in subsequent growing years to maintain or enhance specific morphological traits, such as color and texture, based on changing cultural preferences, sometimes at the cost of reduced cultivability and/or nutrition¹⁶.

Politics and culture have had a strong influence on crop planting and food preferences in ancient (and modern) civilizations. Indeed, in the Andes, previous research showed that under the Inca empire, maize was fulfilled multiple contextual roles. In some cases, it operated as a sacred crop¹⁷, symbolic of political power and tightly associated with high-status or elite social groups^{6, 18}. It was also an important element in numerous cultural events and rituals¹⁹. From an agriculture-economic point-of-view, planting a crop on the most suitable land takes priority if a farmer wishes to maximize the output-to-input ratio. However, in certain civilizations, including Andean groups, maize may not necessarily have been a crop for which yield was considered a deciding factor. For example, in Andean highland communities, maize was not the primary food crop during pre-Inca times²⁰, not only because maize was ill-adapted to arid highland conditions and its cultivation was risky during the short growing season, but also because planting maize did not yield a return on investment. In the Andes during the Spanish

conquest, the chronicler Cobo observed that, “Indians dropped all that they were doing in order to produce a little maize, even though it cost them more than it was worth”. It seemed to him that work in the fields was “...one of the major forms of recreation and festivals...”, and he was impressed by the “extraordinary fondness for farming”²¹, ²². While Indigenous agriculture and foodways would have been far more nuanced than Spanish chroniclers would have appreciated, this supports the fact that Andean farmers were not deterred from planting an abundance of maize in sub-optimal environments. In fact, while it was a desirable food, it could also have acted as a symbol of empire¹⁷ and a ritual-feast crop, since its first introduction to the region around 2500 BCE²³.

The current study contributes to broader investigations documenting maize biogeographic patterns in South America by focusing on maize samples from a 15th century elite stone tomb burial context from south of La Paz, Plurinational State of Bolivia. We sequenced two archaeological Bolivian Maize (aBM) kernels with six libraries, using short read sequencing. In parallel, we used ¹⁴C Accelerator Mass Spectrometer (AMS) radiocarbon dating to determine the approximate age of the samples. Phylogenetic analysis of aBM, 16 archaeological and 226 modern maize samples (*Zea mays* ssp. *Mays* L.), 88 modern samples of Teosinte (*Zea mays* ssp. *parviglumis*), 79 modern samples of Mexicana teosinte (*Zea mays* ssp. *mexicana*) and 1 *Tripsacum* genotype revealed genetic relationships among themselves (Table S1; Dataset S1). Finally, the unique single nucleotide polymorphisms (SNPs) in aBM infer that the selection imposed to enhance reproductive success in marginal highland growing environments resulted in increased maize diversity. Based on the data presented herein, we conclude that a combination the social and political needs of the Inca and local ethnic groups, and mitigation of certain natural environmental factors may have contributed to the specific preference for, and diversification of maize in the Central Andes. As such, our conclusions offer insight into the possible pathways that influenced maize genetic diversity, thus providing a vital resource to investigate further the factors that drove selection for maize domestication and diversity at the cusp of the cultural influence of the Inca empire.

Results

Archaeological Bolivian maize sample (aBM)

La Paz is a major city located in the highlands of west-central modern Bolivia, which is 16°30'S, 68°09'W, 3,650 masl. Dry conditions in this region and the adjacent Andes Mountain range have aided in the preservation of desiccated organic material. Our maize samples were found within a woven camelid fiber pouch that was one of many offerings associated with the mummified remains of a child reported as being found in a *chullpa* (stone tower tomb) south of La Paz, Plurinational State of Bolivia around 1890 (Fig. 1). Although it is impossible to know the exact location of the *chullpa* or the origin of the individual, the assemblage was acquired by the U.S. Consul to Chile and was sent as a gift to the Michigan State University Museum, where she was curated until 2019. Following a series of recent analyses, she has since been voluntarily repatriated to Bolivia²⁴. Given the reported archaeological context of the mummy and her abundant offerings recorded in the MSU Museum intake records, it has been proposed that she might represent an elite member of society. However, the nature of the coarse fabrics, simple design motifs, and restricted color palette of her textiles would suggest a more common individual, perhaps with an affiliation with a local high-status group. The vast majority of the funerary towers that exist in a radius of 60 km south of La Paz city are made of mudbricks. To our knowledge, the only exception is an Inca “cushioned” (i.e., rounded edges) stone and mudbrick tower located in Uypaca, in the nearby Achocalla Valley, 8 km southwest of the city center²⁴. Because of its characteristics, this tomb most likely belonged to a local (possibly non-Inca) high-status group and was part of a more extensive funerary site. Although it is impossible to know the exact provenience of the mummy, it would not be unreasonable to suppose that she originated from this or another similar tomb in the Achocalla Valley. Regardless, in the decades following her

death, community members periodically visited the *chullpa* and left offerings. These included the pouch full of well-preserved maize from which we collected and analyzed samples from two kernels.

A single Accelerator Mass Spectrometer (AMS) date was obtained from one kernel, yielding a radiocarbon age of 400 ± 26 BP (D-AMS-027148; *Zea mays* kernel; corrected for fractionation, no $\delta^{13}\text{C}$ reported). For the date 400 ± 26 BP the two possible calibrated age ranges are 1455–1516 cal AD ($p=0.489$) and 1538–1627 cal AD ($p=0.511$), with a median age of cal AD 1534 (Calibrated at 2σ with Calib 8.2 [SHCal 20.14c]). Choice of the earlier age range is supported by other absolute dates on the assemblage^{25, 26}: 447 ± 45 BP (D-AMS-32518; Cucurbitaceae, $\delta^{13}\text{C} = -27.6$) with two possible age ranges of 1443–1504 cal AD ($p=0.848$) and 1594–1616 cal AD (0.152) with a median age of 1474 cal AD; and 510 ± 20 (PSUAMS-5947; Camelidae fiber, $\delta^{13}\text{C} = -18.9$) calibrated to 1420–1454 cal AD ($p=1.00$) with a median age of 1438 (Both samples calibrated for southern hemisphere at 2σ with Calib 8.2 [SHCal 20.14c]). Ages on the squash and maize are statistically identical at $p=.95$ ($t=1.70$, $\chi^2_{0.05}=3.84$, $df=1$) with a pooled mean of the two ages of 424 ± 18 BP calibrated to 1453–1505 cal AD ($p=0.769$) and 1593–1617 ($p=0.231$). Results on the squash, with highest probability between 1443–1504 cal AD, and camelid fibers with an age range of 1420–1454, as well as the pooled mean, reveal a greatest likelihood for a mid-to late 15th c age for the assemblage, consistent with the earlier of the two calibrated age ranges on the maize kernel.

Two associated maize kernels from the same pouch were subjected to molecular analysis. Genomic base modification patterns associated with ancient DNA (aDNA), including cytosine deamination damage patterns, have been identified. Observed position-specific substitutions at the ends of the sequence reads (e.g., C $\rightarrow$ T/G $\rightarrow$ A^{27–31} (Fig. S1a), as well as an increase in adenine (A) and guanine (G) residues at the 5' end, concomitant with a reduction of C and T residues^{31–33} (Fig. S1b). From this, we evaluated all substitution types and read length (Fig. S2), including read length range³⁴ and substitution rate types, which are associated with aDNA genome data³⁵. In total, these data are consistent with the radiocarbon results, supporting that 1) with the maize is in fact consistent with an archaeological biological sample, and 2) with based on the geographic origin of the sample, the kernels potentially descended from the period of the last Inca empire before the Spaniard Francisco Pizarro's conquest of the Inca capital, Cuzco, in A.D. 1533³⁶.

Our study focused on identifying the location from which aBM originally came, establishing and explaining patterns of genetic variability of maize, with a specific focus on maize strains that are related to our current aBM samples. We were especially interested in exploring how these patterns of variability found in both archaeological and modern samples reflect the influence of cultural systems of behavior on both the spread of variants across the ancient landscape and selection for phenotypic traits by local populations.

Genetic Identity and Genetic Relatedness

After Tupac Inca Yupanqui became the emperor of the Inca Empire in the 15th century, the territory of the Inca empire was greatly expanded across much of western South America, including the Qollasuyu, a broad territory located in what is now western Bolivia, southern Perú, northern Chile, and non-highland Argentina³⁷. Different ethnic groups populated this territory forming polities that were confederated at the time of Inca arrival. Several Aymara Pakaje groups inhabited the highlands and the inter-Andean valleys south of La Paz, from where the maize samples derive³⁸. Because of maize's central role in Inca society—as both a staple crop and a symbolically charged resource among diverse ethnic groups under Inca influence—its widespread distribution may reflect the expansion of Inca social and political hegemony across newly incorporated territories. Given the provenience of the aBM and its age, it is possible the samples were local or alternatively were introduced into western highland Bolivia from the Inca core area – modern Peru. To test this hypothesis, we first evaluated the genetic relatedness of archaeological and contemporary modern maize samples using a genome sequence approach to assist in

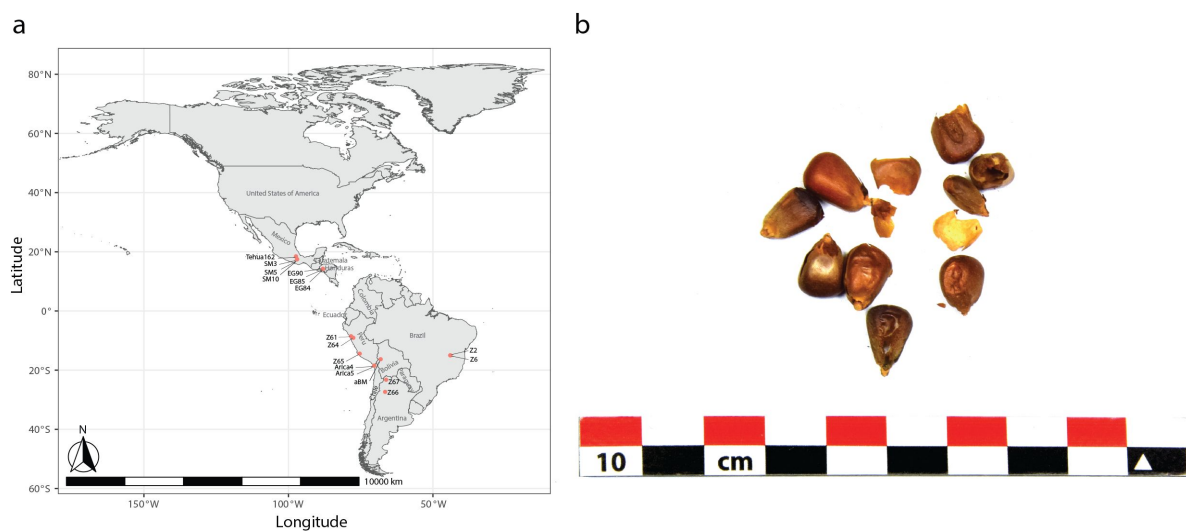


Fig. 1.

Location and appearance of archaeological Bolivian maize (aBM).

a Map of maize sample collection and location information. b Photographs of kernels showing morphological characteristics.

identifying the origin(s) of aBM and the relatedness to other maize from different locations. Our F_3 -statistics results with *Tripsacum* as outgroup suggests that aBM is closely related to two archaeological lowland Peruvian maize (Z65 and Z61) among all other ancient maize and modern maize (**Fig. 2**). These results indicate that the closest relative of aBM is the archaeological maize that came from Peru. In addition, to predict the geographic location of aBM, we used spatial autocorrelations in genetic data, which allowed us to compare it to a set of archaeological maize samples of known geographic origin³⁹. Our results indicate the potential geographic location that aBM may have originally come from (locator.py –seed 549657840) and the predicted location is inside of Peru and very close to archaeological maize Z61 (**Fig. 3**; $R^2(x)=1.0$, $R^2(y)=1.0$). However, this did not indicate that aBM originally came from Inca core area, given that the results are based on predictions and the ages of archaeological samples vary. This suggests that aBM might not only genetically, but also potential geographically related to the archaeological maize from ancient Peru. This is interesting, as historically, descriptions of *capacocha* rituals showed that they were able to source their food from the stations connecting high-altitude ritual sites and other regions of the empire³⁶. Indeed, data generated herein show that our aBM has a strong relationship with archaeological Peruvian maize in terms of both genetic identity and geographic location⁴⁰. This finding is also consistent with the above previous research revealing that one of the maize groups from Peru during Inca peak times spread into newly conquered territories, then slowly diverged to fit the local environment⁴¹ following Inca cultural influence²¹. After this, diverged groups were introduced outside of Inca conquered territories. Although the origin of the aBM – either Inca core area or local – is uncertain, as our results cannot determine the dispersal direction, they do indicate that aBM has a strong relationship with archaeological Peruvian maize.

To investigate the genomic data from the aBM in relation to other archaeological and modern maize, principal component analysis (PCA) was used. As expected, the results showed that all the archaeological maize clustered with our aBM in the same cluster (**Fig. 4**). SM3, 5, and 10 were excluded due to their limited data to meet requirements for the PCA analysis. Unexpectedly, there is one modern maize from Brazil in the ancient maize cluster that has overlap with archaeological Brazilian maize Z6, which indicates that certain genetic traits from ancient Brazilian maize might have been preserved in modern Brazilian maize populations. There is also one modern tropical maize very close to archaeological Peruvian maize Z65. Considering the geographic origin of Z65 and its lowland characters, it is not surprising that modern tropical maize shares some level of genome sequence similarity. Following the establishment of an important maize culture center in Peru, maize was domesticated and diverged into different groups, which then spread along with Inca culture⁴². We found that the Central American cluster has overlap with both North and South America, which supports the distribution route of maize^{13, 43, 44}. This finding also is consistent with previous research revealing that the original maize came from teosinte in Mexico, then dispersed northward into North America and southward into South America^{6, 7, 9, 45, 46}. We also found that a few types of maize from Europe have a much closer distance to the archaeological maize cluster compared to other modern maize, which indicates maize from Europe might expectedly share certain traits or evolutionary characteristics with ancient maize. It is also consistent with the historical fact that maize spread to Europe from multiple areas and time periods after Christopher Columbus's late 15th century voyages to the Americas¹². But as shown, maize also has diversity inside the European maize cluster. It is possible that European farmers and merchants may have favored different phenotypic traits, and the subsequent spread of specific varieties followed the new global geopolitical maps of the Colonial era^{47, 48}.

Taken together, these results support the possibility that aBM was introduced into western highland Bolivia from the Inca core area – modern Peru, and aBM may possibly reveal a history of adaptation to tropical conditions through local agency and/or following cultural influence of the Inca empire^{46, 49}. After this, maize, both as a primary food and as a symbol of power, was dispersed from Peru throughout the Inca empire, in the end enhancing maize diversity in South America.

Fig. 2.

Admixture f3-statistics.

In the form $f_3(\text{aBM}; \text{Test}, \text{tripsacum})$ where Test represents archaeological and modern samples.

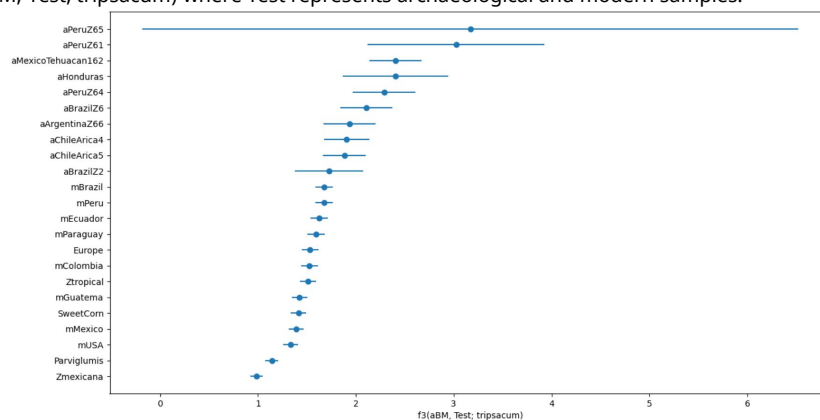
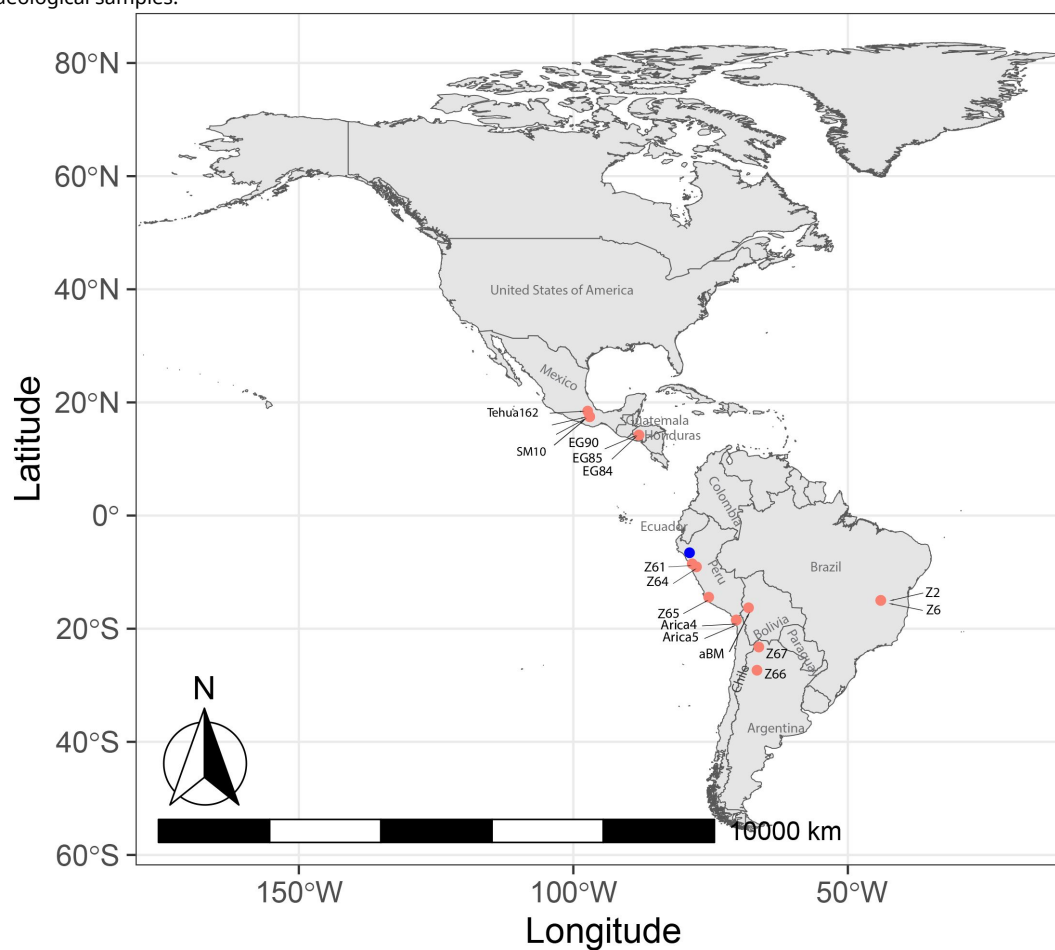


Fig. 3.

Predicted geographic location of archaeological Bolivian maize (aBM).

The blue point is the predicted locations of aBM based on the SNPs from all archaeological maize. The pink points are the archaeological samples.



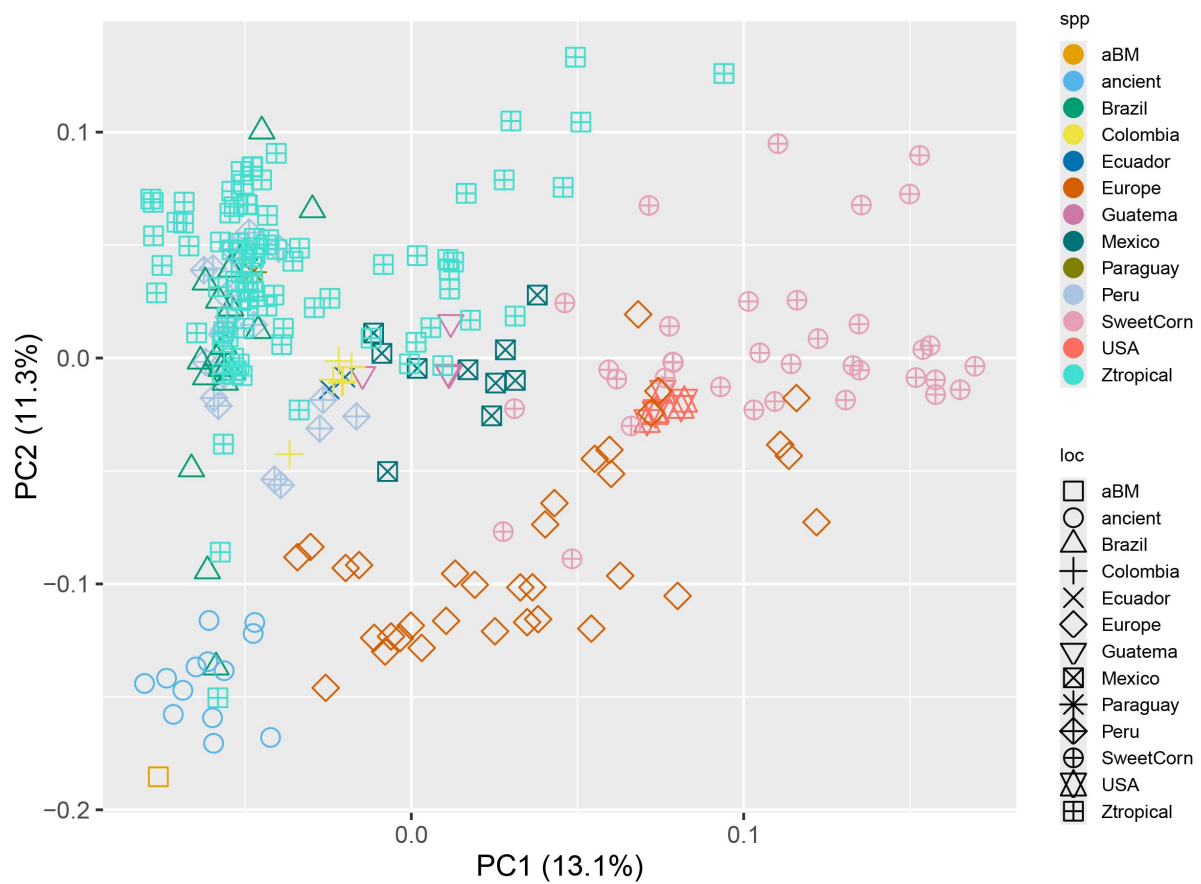


Fig. 4.

Principal Component Analysis (PCA) of nuclear genomic data from archaeological and modern maize.

Samples with greater affinity to one another cluster closer together. Samples with the same color and shape indicate they came from the same country.

Selection for aBM under Inca Culture Influence

After establishing the identity of aBM and its relatedness to other regional samples, we explored how aBM was possibly shaped by human behavior, and specifically how cultural selection within the Inca empire influenced maize diversity in South America. To address this question, we examined the evolutionary relationship between aBM, its wild ancestor, and extant landraces. We constructed a Neighbor-Joining (NJ) and maximum-likelihood (ML) phylogenetic trees using the common genome-wide polymorphisms shared among aBM and modern maize samples⁵⁰. Our data included 226 samples of modern maize, 88 modern samples of Teosinte (*Zea mays* ssp. *parviglumis*), 79 modern samples of Mexican teosinte (*Zea mays* ssp. *mexicana*), and 1 *Tripsacum dactyloides* as an outgroup, totaling 28,588 genome-wide SNPs. As shown in Fig. S3a and Fig. S3b, both phylogenetic trees indicate that aBM lies outside the diversity of modern maize yet clusters within a specific clade alongside one modern tropical maize. This suggests that aBM shares a closer genetic relationship with modern tropical maize while being genetically distinct from other modern maize. Considering modern tropical maize is adapted to tropical climates exhibiting traits such as drought tolerance, heat resistance, and the ability to thrive in low-fertility soils, it is plausible that aBM might share these stress response characteristics with tropical maize^{51, 52}.

To investigate the selection preference on aBM under Inca cultural influence, we compared the allele frequency between aBM and other ancient maize samples. We identified 18,668 SNPs unique to aBM. Notably, two genes -Zm00001eb123120 and Zm00001eb178600-associated with traits “internode length below ear” and “nodes above ear” traits⁵³ contain several unique SNPs in aBM (Dataset S2). This suggested that height related traits in aBM possibly may have been selected under Inca cultural practices. Additionally, variations in the 5' or 3' untranslated regions (UTR) can significantly affect gene expression. We found 83 target genes harboring unique aBM SNPs in their UTR regions (Dataset S3). Gene Ontology (GO) results showed that 3 of these genes are involved in glycosyltransferase activity, while several of others is involved in defense response to biotic and abiotic stress and starch biosynthetic processes (Dataset S4). Previous researches in maize evolution implied that multiple loci are involved in shaping traits variation^{54, 55}. Combining all this information, we inferred those traits associated with plant development and stress response processes might have been selected in maize during the period of Inca cultural influence, although the selection direction is uncertain.

Finally, to understand positive selection in modern maize compared to ancient maize, we investigated the selective sweep using cross-population extended haplotype homozygosity (XP-EHH) method. Following the same method of previous research⁵⁶, the cutoff metric for significant signatures of selection was set at a value of 6 for 5 kb windows across the genome. We only identified one target gene (**Fig. 5**): Zm00001eb105060, whose biological process functions include plant ovule development, chromosome segregation and asparaginyl-tRNA aminoacylation. Ovule development plays a crucial role in improving maize breeding strategies via higher seed production and hybrid vigor⁵⁷. The positive selection of this gene in modern maize suggests that characteristics aiming for higher seed production were selected during maize diversification after aBM. This finding might relate to breeding efforts to improve seed yield in modern maize compared to ancient varieties.

In conclusion, the aBM included in the funerary accoutrements of a young elite Andean girl belongs to the family of ancient Peruvian maize groups, and likely was introduced to the Aymara region following Inca expansion in the late 15th century. Following its arrival in the area of western Bolivia, the Peruvian maize variety was further modified by the local population. Some genomic changes reflect selection to mitigate local environmental conditions, including response to stress. Thus, this case study supports the idea that patterns of regional diversity in pre-Contact South America maize reflect complex biocultural processes. These data may also inform subsequent studies about Colonial dispersals and the origin of global maize varieties¹².

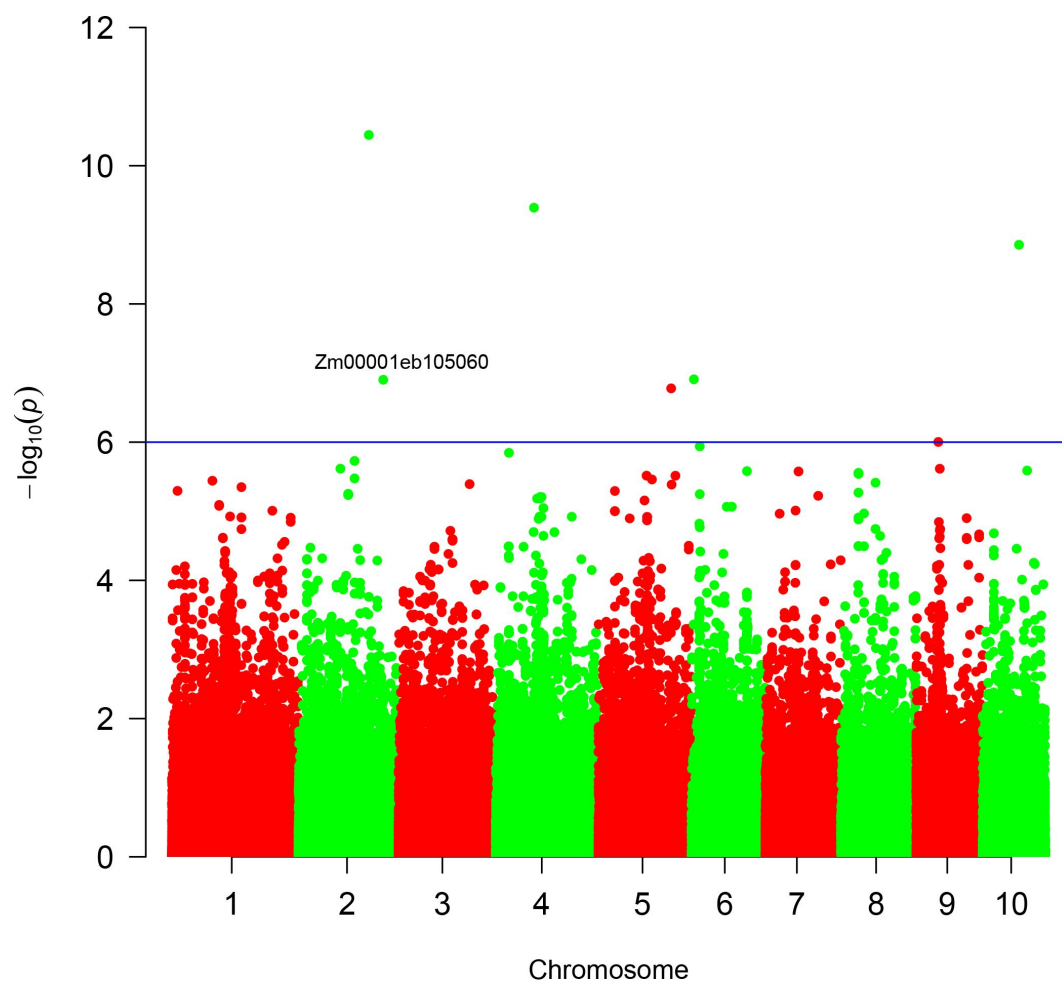


Fig. 5.

Genome-wide distribution of ancient maize specific selective sweeps.

a Signal of selection of modern maize compared to archaeological maize population. Each point indicates a SNP. The blue horizontal line shows the genome-wide significance level ($p = 1 \times 10^{-6}$).

Discussion

Previous archaeological and anthropological studies have demonstrated that cultural and political systems shaped access to and preferences for food. In our study we explored the evidence for the contribution of Inca culture to maize diversity in South America by analyzing 16 archaeological maize samples spanning at least 5,000 years of evolution, and 226 modern maize samples. Recognizing their shared central Andean origins, we propose that the aBM sample potentially came from ancient Peruvian archaeological maize. While our findings are consistent with both genetic and archaeological evidence, they raise several questions. Notably, what specific traits were selected in aBM, and how this selection occurred under Inca cultural practices is unclear, especially given our limited aBM sample size and the lack of the same genotype in archaeological maize under non-Inca culture influence.

Cultural preferences are often a factor in maintaining certain aesthetically desirable traits¹⁴,¹⁵, such as color and texture, even at the cost of reduced cultivability or nutrition¹⁶. This suggests it is possible that culture played an important role in maize evolution and diversity. To better understand how exactly cultural interactions influenced maize diversity in the Central Andes, we explored the possible origin of aBM and its closest genetic relationships to ancient maize. We identified that aBM potentially have originated from ancient Peru, and archaeological maize from Peru has the closest genetic relationship with aBM, indicating that aBM may be a product of Inca cultural dissemination into new territories²¹.

We further investigated which traits were possibly selected under Inca culture by analyzing unique SNPs in aBM. The results show that traits related to plant development, and stress response were selected. These traits can aid in resisting drought and frost in the relatively harsh conditions of the Andean highlands. The selected traits could facilitate fermentation for the creation of *chicha* to meet the consumption requirement of elite lifeways¹⁸. However, we are uncertain if these traits were directly selected under Inca culture due to the limited aBM sample size, which prevented us from identifying positive selection directly between aBM and other archaeological maize groups.

Traditional breeding systems have increased tolerance to biotic and abiotic stress and contributed to yield in crops⁵⁸,⁵⁹. In this study, we compared modern maize to archaeological maize across the entire Western Hemisphere and found that the ovule development traits were selected for breeding to improve maize seed production during its evolution. However, it is uncertain whether this selection happened directly in modern maize when compared to ancient maize, or if it happened during the process of modern improvement.

Our results not only highlight the influence of culture on maize diversity in South America but also point to the adaptive characteristics of maize in improving maize seed production during evolution. This provides a potential source for maize diversity studies and breeding programs. Most importantly, this work suggests that cultural influence plays an important role in aiding maize diversity, alongside the traits selected by early farmers.

Materials and Methods

Plant material

The two archaeological Bolivian maize (aBM) kernels were collected prior to 1890 AD from the area south of modern La Paz, Bolivia, and later curated at the Michigan State University Museum until their repatriation in 2022²⁵. AMS assay dates reveal a mid-late 15th century date for the maize samples. In addition, data from 16 archaeological samples collected from previous published research were also used, including two from Chile dating from 750 to 1,100 BP, two from Brazil dating from 510 to 690 BP, three from Peru dating from 600 to 1,000 BP, two from Argentina, dating from 980 to 1,040 BP and from 70 to 130 BP, three from Honduras dating from 1,740 to 2,300 BP, and three from Mexico dating to 5,310 BP (Table S1)^{40, 60}.

aDNA extraction

Archaeological maize kernels were processed in a clean laboratory environment at Michigan State University. Whole kernels were crushed in a mortar and pestle, to which 600 uL of fresh PTB (*N*-phenacylthiazolium bromide) buffer (0.4 mg/mL proteinase K, 2.5 mM PTB, 50 nM dithiothreitol) was added. The sample was then added to 1260 uL of lysis buffer. An additional 600 uL of PTB buffer was used to rinse the mortar and added to the lysis buffer. The individually ground maize kernels were rotated overnight at 37 °C. The remaining DNA extraction followed Swartz et al., 2017¹³.

Library construction, amplification, and sequence

DNA extracts were converted to Illumina-compatible paired-end DNA libraries in a clean setting using SMARTer ThurPLEX DNA-seq from Takara Bio and amplified with 7 polymerase chain reactions (PCR) cycles. The generated libraries were cleaned to remove remaining adaptors with AxyPrep Mag beads (Fisher Scientific) and analyzed on a 2100 Bioanalyzer (Agilent Technologies) to check for quality. The libraries were indexed to facilitate pooling on the NovaSeq SP at the University of Illinois (Urbana-Champaign).

Geographic location map

The geographical coordinates of each sample were obtained from previously published studies, as indicated (Table S1). Utilizing latitude and longitude information, the samples were plotted on a geographic map, resulting in the successful mapping of 17 ancient samples. Geographic data visualization was carried out in R using the ggplot2, sf, leaflet, rnatuarearth, rnatuarearthdata, ggspatial, and rnatuarearthhires packages.

Genome analysis

For the aBM sequence samples data with six libraries, we combined them into a single composite sample, referred to as aBMComb. This was achieved by mapping each individual aBM sequence sample to the B73 V5 soft-masked maize genome and subsequently removing duplicate reads before merging them. We merged the sequence sample to obtain higher coverage for the analysis was needed for this study.

Subsequently, we used MapDamage v.2.2.1⁶¹ verified cytosine deamination profiles consistent with ancient DNA, and estimated the misincorporation frequencies. Based on these results we hard-masked 5' thymine and 3' adenine residues within 5nt of the two ends, where deaminations was most concentrated.

Data for 16 ancient samples with specific geographic location information (maize landraces) obtained from previously published work^{40, 60, 62} by downloading sequencing from Sequence Read Archive (SRA) (<https://www.ncbi.nlm.nih.gov/sra/>)⁶³. To ensure consistency and accuracy, we downloaded the raw sequence (SRA format) for each sample and converted it into FASTQ format using fastq-dump.

For the archaeological maize samples, adapters were removed and paired reads were merged using AdapterRemoval⁶⁴ with parameters `--minquality 20 --minlength 30`. All 5′ thymine and 3′ adenine residues within 5nt of the two ends were hard-masked, where deamination was most concentrated. Reads were then mapped to soft-masked B73 v5 reference genome using BWA v0.7.17 with disabled seed (`-l 1024 -o 0 -E 3`) and a quality control threshold (`-q 20`) based on the recommended parameter⁶⁵ to improve ancient DNA mapping. Duplicated reads were removed by GATK tools Picard (<https://gatk.broadinstitute.org/hc/en-us>). The coverage of each nucleotide position was estimated by bam-readcount (<https://github.com/genome/bam-readcount>) (Table S2), and the overall coverage of the assembled genome sequence was measured using Qualimap v2.2.1⁶⁶.

Variant Calling

We used SequenceTools pileupCaller (v 1.5.4.0; <https://github.com/stschiff/sequenceTools>) to generate pseudohaploid calls for 17 archaeological maize samples. The commands executed were: `samtools mpileup -R -B -q20 -Q20 & pileupCaller -- sampleNames <ancient sample name> randomHaploid -- singleStrandMode`. This process yielded 16,815,782 single nucleotide polymorphisms (SNPs).

We further filtered the SNPs using estimated information from bam-readcount. SNPs supported by at least two reads were retained for each individual archaeological maize sample. We then combined the filtered SNPs from each sample, keeping only those SNPs that were present in at least one sample. This resulted in a final dataset of 2,808,917 SNPs for the 17 archaeological maize samples.

For all subsequent analyses, the ancient genotype data was merged with the genotype dataset for 394 samples from Grzybowski et al, (“Feb 03, 2023 version files” in Dryad repository: <https://doi.org/10.5061/dryad.bnzs7h4f1>). The merged dataset included 2,635,047 SNPs from chromosomes 1-10 that were common in both archaeological and modern maize samples, which were used for downstream analysis.

*f*₃ statistic and phylogenetic analyses

We created a genotype file by combining 17 archaeological maize and publicly available SNP dataset from Grzybowski et al, resulting in a total of 17-archaeological and 226-modern maize, 88-modern samples of Teosinte (*Zea mays* ssp. *parviglumis*), 79-modern samples of *Mexicana teosinte* (*Zea mays* ssp. *mexicana*), and 1-*Tripsacum* genotypes.

Next, we used plink v1.9⁶⁷ to prune for linkage disequilibrium with the parameters `--indep-pairwise 50 5 0.1`, following previous research⁴⁵, to prepare the dataset for Principal Component Analysis (PCA) using “plink -pca”. The final results were plotted in R using the ggplot2 package⁶⁸.

We calculated the *f*₃ statistic^{69, 70} using the admixr package⁷¹, which is part of the package of ADMIXTURE software suite⁷². The outgroup *f*₃ statistics was used to measure how closely aBM and each test group are related to each other. The final results were plotted using Python.

A total of 28,588 SNPs present in both aBM and all modern maize, out of a total of 2,626,087 SNPs, was used to build the phylogenetic tree. A maximum likelihood (ML) phylogenetic tree was constructed using SNPhylo⁷³ with parameter: `-v snps.vcf -r -o tripsacum -P tree -b 1000`; A Neighbour-Joining (NJ) phylogenetic tree was constructed using VCF2PopTree⁷⁴ with the “number of differences” model. Both ML and NJ phylogenies trees were inferred and visualized using FigTree v1.4.4 (<http://tree.bio.ed.ac.uk/software/figtree/>).

Geographical location prediction

To predict the possible geographical location of aBM, we applied a deep neural network method³⁹ using only 17 ancient maize samples, as all the archaeological maize clustered in the same cluster. A total of 2,800,190 SNPs were used to predict the possible geographical location of aBM. The command used was: `locator.py --vcf SNP_file --sample_data sample_data.txt --out name`.

Infer selection on aBM

Due to the lack of archaeological maize with the same genotype as aBM grows outside the influence of Inca culture, it was impossible to directly identify the selection that occurred on aBM under Inca cultural influence. To determine if any of our SNPs are related to known traits, we utilized existing information on SNP-trait associations⁵³. To achieve potential unique selection on aBM, we calculated the allele frequency for each SNPs between aBM and other ancient maize, resulting in allele frequency data for 49,896 SNPs. Of these, 18,668 SNPs were unique to aBM. We consulted SNPVersion 2.0 from MaizeGDB^{50, 75} and found that 84 of these unique SNPs were classified as “3_prime_UTR_variant” or “5_prime_UTR_variant”. These SNPs corresponded to 83 target genes, for which we retrieved Gene Ontology (GO) terms.

Identification of selection sweeps

To detect genomic signatures of recent selective sweeps in modern maize compared to ancient maize groups, SNPs that present in at least two archaeological samples and modern samples were used. We applied the cross-population extended haplotype homozygosity (XP-EHH)⁷⁶ method. A commonly used threshold $-\log_{10}(p) = 6$ was set as the cutoff threshold for identifying the signal of significant selection. The regions with $-\log_{10}(p) \geq 6$ were designated as candidate sweeps for candidate genes located within 5 KB of these regions identified, following the approach used in previous research⁵⁶. MaizeMine from MaizeGDB⁷⁷ was used to obtain putative gene function information.

Acknowledgements

We are grateful to all participants who were involved in this project. We are thankful for the support of the Michigan State University Cloud Fellowship for computational skills training. This work was supported in part through computational resources and services provided by the Institute for Cyber-Enabled Research at Michigan State University. We thank Jennifer Wai, University of Chicago, for her help with DNA extraction and library preparation. We also thank Emily Josephs, Michigan State University, and Logan Kistler, Smithsonian Institution of National Museum of Natural History, for their help with the DNA analysis. Thanks to José Capriles Flores for his insightful reading and comments. The MSU Museum provided maize samples and images for documentation and analysis prior to repatriation to Bolivia. Research in the laboratory of Brad Day was supported by the MSU Research Foundation.

Additional information

Author Contributions

HC, ABY, AT, and BD designed and conceived the experiments. HC, ABY, AT, BD, TM, SW, CRC, WL, and GW analyzed the data and wrote the manuscript. All authors commented and agreed on the manuscript before submission.

Data availability

The authors declare that all data supporting the findings of this study are included in the manuscript and its supplementary files or are available from the corresponding author upon request. Source data are provided with this paper. The genome sequence data of ancient Bolivian Maize has been deposited in the NCBI database under BioProject accession PRJNA886637. Custom scripts and single nucleotide polymorphism (SNP) calls are available on Dryad (DOI: <https://doi.org/10.5061/dryad.w6m905qtd> )

Additional files

Supplemental Figure 1 

Supplemental Figure 2 

Supplemental Figure 3 

Supplemental Figure 4 

Supplemental Dataset S1 

Supplemental Dataset S2 

Supplemental Dataset S3 

Supplemental Tables 

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Reviewer #1 (Public review):

Summary:

In this manuscript, authors describe a good quality ancient maize genome from 15th century Bolivia and try to link the genome characteristics to Inca influence. Overall, the revised manuscript is still below the standard in the field. While dating of the sample and the authentication of ancient DNA has been evidenced robustly, the downstream genetic analyses do not support the conclusion that genomic changes can be attributed to Inca influence.

There is more story telling than story testing in this manuscript, analyses are not robust and possibly of very narrow interest.

Strengths:

Technical data related to the maize sample are robust. Radiocarbon dating strongly evidenced sample age, estimated to around 1474 AD. Authentication of ancient DNA has been done robustly. Spontaneous C-to-T substitutions which are present in all ancient DNA are visible in reported sample with the expected pattern. Despite low fraction of C-to-T at the 1st base, this number could be consistent with cool and dry climate in which the sample was preserved. The distribution of DNA fragment sizes is consistent with expectations for sample of this age.

Weaknesses:

(1) The geographic placement of the sample based on genetic data is not robust. To make use of the method correctly, it would be necessary to validate that genetic samples in this region follow the assumption of the 'isolation-by-distance' with dense sampling, which has not been done. Without this important information, we do not know if genetic similarity is influenced by demographic events and/or selection. The analysis is not a robust evidence of sample connectivity.

(2) The conclusion that Ancient Andean maize is genetically similar to European varieties and hence share similar evolutionary history is not well supported. PCA plot in Fig. 4 merely represents sample similarity based on two components (jointly responsible for about 20% of variation explained). Contrary to authors' conclusion, the direct test of similarity using outgroup F_3 statistic does not support that European varieties are particularly closely related to ancient Andean maize. These levels of shared drift could be due ancient Andean maize relationship with other related groups, such as ancient or modern Brazil. A relationship test between multiple populations would be necessary to show significant direct relationship between ancient Andean maize and European maize.

(3) The conclusion that selection detected in aBM sample is due to Inca influence has no support. Firstly, selection signature can be due to environmental or any other factors. To disentangle those, authors would need to generate the data for a large number of samples from similar cultural context and from a wide-ranging environmental context followed by a formal statistical test. Secondly, allele frequency increase can be attributed to selection or demographic processes, and alone is not a sufficient evidence for selection. Presented XP-EHH method seems unsuitable for single individual. Overall, methods used in this paper raise some concerns: i) how accurate are allele-frequency tests of selection when only single individual is used as a proxy for a whole population, ii) the significance threshold has been arbitrary fixed to an absolute number based on other studies, but the standard is to use, for example, top fifth percentile.

In sum, this manuscript presents new data that seem to be of high quality, but the analyses are frequently inappropriate and/or over-interpreted.

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Reviewer #2 (Public review):

I am glad to see a revised version of the manuscript. The authors have successfully handled some of my comments, but others require additional attention. In particular, the dataset seems quite robust and valuable to publish, and the descriptive analysis of its position relative to other modern and ancient genomes is generally sound. The selection analyses

remain unsupported, and should be removed from the paper. In addition, I agree with the other reviewers and reiterate my comment that the Locator analysis is not robust.

As I said in my original review, the XP-EHH method is not applicable to pseudohaploid variant calls in a single individual. This method is simply not appropriate to apply to the data at hand, as the method relies on knowledge of diploid genotypes, usually phased, and the results from this test are not robust. It is possible that the XP-EHH method could be extended to this data type or genotype likelihoods with extensive validation and conditioning on a large reference panel, but in general haplotype-based approaches have not been extensible to low-coverage pseudohaplotype datasets. At any rate, any off-the-shelf implementation is inappropriate and unsupported. I am sorry to be this negative about this analysis, but it cannot be used as presented, the results from using it in this way would be spurious by definition.

In addition, identifying GO terms without statistical assessment of enrichment is not a robust analysis, nor is selecting genes with a high proportion of rare alleles without extensive additional contextualization based on the expectations of neutrality and deviations potentially tied to selection. For this reason, the two genes linked with height traits have no support here as genuinely being targets of selection. It is a frustrating reality for us in the ancient DNA field that small numbers of highly degraded genomes offer extremely limited scope for selection analyses, but that's the unfortunate state of play, and is the situation here.

My other major critique remains the application of the Locator method. As Reviewer 1 notes, this method must be built on a densely sampled dataset with strong isolation by distance, which is not done here. The authors explained their approach with more detail in their response, but it is fundamentally inappropriate for this dataset. It does not add anything more than the f3 analysis, and creates a falsely precise inference of genetic-geographic origins that is not supported.

Per authors' response to my previous recommendation 6, it is not advisable to re-map the reads after damage masking, and doing this with a conservative hard-masking approach will lead to a high mismatch rate and significant loss of reads in BWA. This could also exacerbate reference sequence bias which is already a major challenge for ancient DNA (see Gunther et al 2019 PLoS Genet). The correct approach is to map reads, mask or rescale for damage, and then proceed with the modified alignment file. In response to Reviewer 3's comment 3, the authors also refer to a "0 mismatch alignment" strategy. This is not concordant with the damage analysis, and if they truly do not allow mismatches this would be very inadvisable, as it would allow an extreme reference sequence bias.

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Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Summary:

In this manuscript, the authors describe a good-quality ancient maize genome from 15th-century Bolivia and try to link the genome characteristics to Inca influence. Overall, the manuscript is below the standard in the field. In particular, the geographic origin of the sample and its archaeological context is not well evidenced. While dating of the sample and the authentication of ancient DNA have been evidenced robustly, the downstream genetic analyses do not support the conclusion that genomic changes can

be attributed to Inca influence. Furthermore, sections of the manuscript are written incoherently and with logical mistakes. In its current form, this paper is not robust and possibly of very narrow interest.

Strengths:

Technical data related to the maize sample are robust. Radiocarbon dating strongly evidenced the sample age, estimated to be around 1474 AD. Authentication of ancient DNA has been done robustly. Spontaneous C-to-T substitutions, which are present in all ancient DNA, are visible in the reported sample with the expected pattern. Despite a low fraction of C-to-T at the 1st base, this number could be consistent with the cool and dry climate in which the sample was preserved. The distribution of DNA fragment sizes is consistent with expectations for a sample of this age.

Weaknesses:

Thank you for all your thoughtful comments. See below for comments on each.

(1) Archaeological context for the maize sample is weakly supported by speculation about the origin and has unreasonable claims weighing on it. Perhaps those findings would be more convincing if the authors were to present evidence that supports their conclusions: i) a map of all known tombs near La Paz, ii) evidence supporting the stone tomb origins of this assemblage, and iii) evidence supporting non-Inca provenance of the tomb.

We believe we are clear about what information we have about context. First, the intake records from the MSU Museum from 1890 are not as detailed as we would like, but we cannot enhance them. The mummified girl and her accoutrements, including the maize, came from a stone tower or chullpa south of La Paz, in what is now Bolivia. We do not know which stone chullpa, so a map would be of limited use. The mortuary group is identified as Inca, but as we note the accoutrements do not appear of high status, so it is possible that she is not an elite. Mud tombs are normally attributed to the local population, and stone towers to Inca or elites. We have clarified at multiple places in the text that the maize is from the period of Inca incursion in this part of Bolivia and have modified text to reflect greater uncertainty of Inca or local origin, but that selection for environmentally favorable characteristics had taken place. Regardless, there are three 15th c CE or AD AMS ages on the maize, a cucurbita rind, and a camelid fiber. The maize is almost certainly mid to late 15th century CE.

(2) Dismissal of the admixture in the reported samples is not evidenced correctly. Population f_3 statistic with an outgroup is indeed one of the most robust metrics for sample relatedness; however, it should not be used as a test of admixture. For an admixture test, the population f_3 statistic should be used in the form: i) target population, ii) one possible parental population, iii) another possible parental population. This is typically done iteratively with all combinations of possible parental populations. Even in such a form, the population f_3 statistic is not very sensitive to admixture in cases of strong genetic drift, and instead population f_4 statistic (with an outgroup) is a recommended test for admixture.

We have removed “Our admixture f_3 -statistics test results suggest aBM is not admixed” in our revised manuscript. Since our goal here is to identify which group(s) has(have) the highest relatedness with aBM, so population f_3 statistic with an outgroup is the most robust metric to do the test and to support our conclusion here.

(3) The geographic placement of the sample based on genetic data is not robust. To make use of the method correctly, it would be necessary to validate that genetic samples

in this region follow the assumption of the 'isolation-by-distance' with dense sampling, which has not been done. Additionally, the authors posit that "This suggests that aBM might not only be genetically related to the archaeological maize from ancient Peru, but also in the possible geographic location." The method used to infer the location is based on pure genetic estimation. The above conclusion is not supported by this method, and it directly contradicts the authors' suggestion that the sample comes from Bolivia.

We understood that it is necessary to validate the assumption of the 'isolation-by-distance' with dense sampling. But we did not do it because: 1) the ancient maize age ranges from ~5000BP to ~100BP and they were found in very different countries at different times. 2) isolation-by-distance is a population genetic concept and it's often used to test whether populations that are geographically farther apart are also more genetically different. Considering we only have 17 ancient samples in total our sample size is not sufficient for a big population test.

For "It directly contradicts the authors' suggestion that the sample comes from Bolivia.", as we described in our manuscript that "Given the provenience of the aBM and its age, it is possible the samples were local or alternatively were introduced into western highland Bolivia from the Inca core area – modern Peru." The sample recording file did show the aBM sample was found in Bolivia, but we do not know where aBM originally came from before it was found in Bolivia. To answer this question, we used locator.py to predict the potential geographic location that aBM may have originally come from, and our results showed that the predicted location is inside of modern Peru and is also very close to archaeological Peruvian maize.

Therefore, our conclusion that "This suggests that aBM might not only be genetically related to the archaeological maize from ancient Peru, but also in the possible geographic location" does not contradict that the sample was found Bolivia.

(4) The conclusion that Ancient Andean maize is genetically similar to European varieties and hence shares a similar evolutionary history is not well supported. The PCA plot in Figure 4 merely represents sample similarity based on two components (jointly responsible for about 20% of the variation explained), and European samples could be very distant based on other components. Indeed, the direct test using the outgroup f_3 statistic does not support that European varieties are particularly closely related to ancient Andean maize. Perhaps these are more closely related to Brazil? We do not know, as this has not been measured.

Our conclusion is "We also found that a few types of maize from Europe have a much closer distance to the archaeological maize cluster compared to other modern maize, which indicates maize from Europe might expectedly share certain traits or evolutionary characteristics with ancient maize. It is also consistent with the historical fact that maize spread to Europe after Christopher Columbus's late 15th century voyages to the Americas. But as shown, maize also has diversity inside the European maize cluster. It is possible that European farmers and merchants may have favored different phenotypic traits, and the subsequent spread of specific varieties followed the new global geopolitical maps of the Colonial era".

We understood your concerns that two components only explain about 20% of the variation. But as you can see from the Figure 2b in Grzybowski, M.W. et al., 2023 publication, it described that "the first principal component (PC1) of variation for genetic marker data roughly corresponded to the division between domesticated maize and maize wild relatives is only 1.3%". It shows this is quite common in maize, especially when the datasets include landraces, hybrids, and wild relatives. For our maize dataset, we have archaeological maize data ranging from ~5,000BP to ~100BP, and we also have modern maize, which makes the

genetic structure of our data more complicated. Therefore, we think our two components are currently the best explanation currently possible. We also included PCA plot based on component 1 and 3 in Fig4_PCA13.pdf. It does not show that the European samples are very distant.

For “Perhaps these are more closely related to Brazil?”, thank you for this very good question, but we apologize that we cannot answer this question from our current study because our study focuses on identifying the location where aBM originally came from, establishing and explaining patterns of genetic variability of maize, with a specific focus on maize strains that are related to our current aBM. Thus, we will not explore the story between maize from Brazil and European maize in our current study.

(5) The conclusion that long branches in the phylogenetic tree are due to selection under local adaptation has no evidence. Long branches could be the result of missing data, nucleotide misincorporations, genetic drift, or simply due to the inability of phylogenetic trees to model complex population-level relationships such as admixture or incomplete lineage sorting. Additionally, captions to Figure S3, do not explain colour-coding.

We have removed “aBM tends to have long branches compare to tropicalis maize, which can be explained by adaption for specific local environment by time.” in our revised manuscript.

We have added the color-coding information under Fig. S3 in our revised manuscript.

(6) The conclusion that selection detected in aBM sample is due to Inca influence has no support. Firstly, selection signature can be due to environmental or other factors. To disentangle those, the authors would need to generate the data for a large number of samples from similar cultural contexts and from a wide-ranging environmental context, followed by a formal statistical test. Secondly, allele frequency increase can be attributed to selection or demographic processes, and alone is not sufficient evidence for selection. The presented XP-EHH method seems more suitable. Overall, methods used in this paper raise some concerns: i) how accurate are allele-frequency tests of selection when only single individual is used as a proxy for a whole population, ii) the significance threshold has been arbitrary fixed to an absolute number based on other studies, but the standard is to use, for example, top fifth percentile. Finally, linking selection to particular GO terms is not strong evidence, as correlation does not imply causation, and links are unclear anyway.

In sum, this manuscript presents new data that seems to be of high quality, but the analyses are frequently inappropriate and/or over-interpreted.

Regarding your suggestion that “from similar cultural contexts and from a wide-ranging environmental context, followed by a formal statistical test”, we apologize that this cannot be done in our current study because we could not find other archaeological maize samples/datasets that are from similar cultural contexts.

For “Secondly, allele frequency increase can be attributed to selection or demographic processes, and alone is not sufficient evidence for selection.” Yes, we agree, and that’s why we said it “inferred” the conclusion instead of “indicated”. Furthermore, we revised the whole manuscript following all reviewers’ comments and reorganized and reduced the part on selection on aBM.

For “The presented XP-EHH method seems more suitable”, we do not think XP-EHH is the best method that could be used here because we only have one aBM sample, but XP-EHH is more suitable for a population analysis.

For “Finally, linking selection to particular GO terms is not strong evidence, as correlation does not imply causation, and links are unclear anyway.”, as we described in our manuscript, our results “inferred” instead of “indicated” the conclusion.

Reviewer #2 (Public review):

Summary:

The manuscript presents valuable new datasets from two ancient maize seeds that contribute to our growing understanding of the maize evolution and biodiversity landscape in pre-colonial South America. Some of the analyses are robust, but the selection elements are not supported.

Strengths:

The data collection is robust, and the data appear to be of sufficiently high quality to carry out some interesting analytical procedures. The central finding that aBM maize is closely related to maize from the core Inca region is well supported, although the directionality of dispersal is not supported.

Weaknesses:

Thank you for your comments and suggestions. See below for responses and explanations.

The selection results are not justified, see examples in the detailed comments below.

(1) The manuscript mentions cultural and natural selection (line 76), but then only gives a couple of examples of selecting for culinary/use traits. There are many examples of selection to tolerate diverse environments that could be relevant for this discussion, if desired.

We have added related examples with references supported in our revised manuscript.

(2) I would be extremely cautious about interpreting the observations of a Spanish colonizer (lines 95-99) without very significant caveats. Indigenous agriculture and food ways would have been far more nuanced than what could be captured in this context, and the genocidal activities of the Europeans would have impacted food production activities to a degree, and any contemporaneous accounts need to be understood through that lens.

We agree with the first part of this comment and have softened our use of this particular textual material such that it is far less central to interpretation. While of interest, we cannot evaluate the impact of colonial European activities or observational bias for purposes of this analysis.

(3) The f_3 stats presented in Figure 2 are not set up to test any specific admixture scenarios, so it is unsupported to conclude that the aBM maize is not admixed on this basis (lines 201-202). The original f_3 publication (Patterson et al, 2012) describes some scenarios where f_3 characteristics associate with admixture, but in general, there are many caveats to this approach, and it's not the ideal tool for admixture testing, compared with e.g., f_4 and D (abba-baba) statistics.

You make an important point that f_3 stats is not the ideal tool for admixture testing. Since our study goal here is to identify which group(s) has(have) the highest relatedness with aBM, the population f_3 statistic with an outgroup is the most robust metrics with which to do the test

and to support our conclusion here. We have removed the “Our admixture f3-statistics test results suggest aBM is not admixed” in our revised manuscript.

(4) I'm a little bit skeptical that the Locator method adds value here, given the small training sample size and the wide geographic spread and genetic diversity of the ancient samples that include Central America. The paper describing that method (Battey et al 2020 eLife) uses much larger datasets, and while the authors do not specifically advise on sample sizes, they caution about small sample size issues. We have already seen that the ancient Peruvian maize has the most shared drift with aBM maize on the basis of the f3 stats, and the Locator analysis seems to just be reiterating that. I would advise against putting any additional weight on the Locator results as far as geographic origins, and personally I would skip this analysis in this case.

As we described in our manuscript, we have 17 archaeological samples in total. Please find more detailed information from the “geographical location prediction” section.

We cannot add more ancient samples because they are all that we could find from all previous publications. We may still want to keep this analysis because f3 stats indicates the genome similarity, but the purpose of locator.py analysis is indicating the predicted location of origin of a genetic sample by comparing it to a set of samples of known geographic origin.

(5) The overlap in PCA should not be used to confirm that aBM is authentically ancient, because with proper data handling, PCA placement should be agnostic to modern/ancient status (see lines 224-226). It is somewhat unexpected that the ancient Tehuacan maize (with a major teosinte genomic component) falls near the ancient South American maize, but this could be an artifact of sampling throughout the PCA and the lack of teosinte samples that might attract that individual.

We have removed “which supports the authenticity of aBM as archaeological maize” in our revised manuscript. The PCA was only applied for all maize samples, so we did not include any teosinte samples in the analysis.

(6) What has been established (lines 250-251) is genetic similarity to the Inca core area, not necessarily the directionality. Might aBM have been part of a cultural region supplying maize to the Inca core region, for example? Without a specific test of dispersal directionality, which I don't think is possible with the data at hand, this is somewhat speculative.

We added this and re-wrote this part in our revised manuscript.

(7) Singleton SNPs are not a typical criterion for identifying selection; this method needs some citations supporting the exact approach and validation against neutral expectations (line 278). Without Datasets S2 and S3, which are not included with this submission, it is difficult to assess this result further. However, it is very unexpected that ~18,000 out of ~49,000 SNPs would be unique to the aBM lineage. This most likely reflects some data artifact (unaccounted damage, paralogs not treated for high coverage, which are extremely prevalent in maize, etc). I'm confused about unique SNPs in this context. How can they be unique to the aBM lineage if the SNPs used overlap the Grzybowski set? The GO results do not include any details of the exact method used or a statistical assessment of the results. It is not clear if the GO terms noted are statistically enriched.

We have added references 53 and 54 in our revised manuscript, and we also uploaded the Datasets S2 and S3.

For “I’m confused about unique SNPs in this context. How can they be unique to the aBM lineage if the SNPs used overlap the Grzybowski set?”, as we described in our materials and method part that “To achieve potential unique selection on aBM, we calculated the allele frequency for each SNPs between aBM and other archaeological maize, resulting in allele frequency data for 49,896 SNPs. Of these, 18,668 SNPs were unique to aBM.” Thus, the unique SNPs for aBM came from the comparison between aBM with other archaeological maize, and we did not use any modern maize data from the Grzybowski set.

For “The GO results do not include any details of the exact method used or a statistical assessment of the results. It is not clear if the GO terms noted are statistically enriched.” We did not do GO Term enrichment, so there are no statistical assessments for the results. What we have done was we retained the GO Terms information for each gene by checking their biological process from MaizeGDB, after that, we summarized the results in Dataset S4.

(8) The use of XP-EHH with pseudo haplotype variant calls is not viable (line 293). It is not clear what exact implementation of XP-EHH was used, but this method generally relies on phased or sometimes unphased diploid genotype calls to observe shared haplotypes, and some minimum population size to derive statistical power. No implementation of XP-EHH to my knowledge is appropriate for application to this kind of dataset.

We used the same XP-EHH as this publication “Sabeti, P.C. et al. Genome-wide detection and characterization of positive selection in human populations. *Nature* 449, 913-918 (2007).” Specifically in our analysis, the SNP information of modern maize was compared with ancient maize. The code is available in <https://doi.org/10.5061/dryad.w6m905qtd>.

XP-EHH is a statistical method used in population genetics to detect recent positive selection in one population compared to another, and it often applied in modern large maize populations in previous research. In our study, we wanted to detect recent positive selection in modern maize compared to ancient maize, thus, we applied XP-EHH here. Although the population size of ancient maize is not big, it is the best method that we can apply for our dataset here to detect recent selection on modern maize.

Reviewer #3 (Public review):

Summary:

The authors seek to place archaeological maize samples (2 kernels) from Bolivia into genetic and geographical context and to assess signatures of selection. The kernels were dated to the end of the Incan empire, just prior to European colonization. Genetic data and analyses were used to characterize the distance from other ancient and modern maize samples and to predict the origin of the sample, which was discovered in a tomb near La Paz, Bolivia. Given the conquest of this region by the Incan empire, it is possible that the sample could be genetically similar to populations of maize in Peru, the center of the Incan empire. Signatures of selection in the sample could help reveal various environmental variables and cultural preferences that shaped maize genetic diversity in this region at that time.

Strengths:

The authors have generated substantial genetic data from these archaeological samples and have assembled a data set of published archaeological and modern maize samples that should help to place these samples in context. The samples are dated to an interesting time in the history of South America during a period of expansion of the Incan empire and just prior to European colonization. Much could be learned from even this small set of samples.

Weaknesses:

Many thanks for your comments and suggestions. We have addressed these below and provided further explanation.

(1) Sample preparation and sequencing:

Details of the quality of the samples, including the percentage of endogenous DNA are missing from the methods. The low percentage of mapped reads suggests endogenous DNA was low, and this would be useful to characterize more fully. Morphological assessment of the samples and comparison to morphological data from other maize varieties is also missing. It appears that the two kernels were ground separately and that DNA was isolated separately, but data were ultimately pooled across these genetically distinct individuals for analysis. Pooling would violate assumptions of downstream analysis, which included genetic comparison to single archaeological and modern individuals.

We did not do the morphological assessment of the samples and comparison to morphological data from other maize varieties because we only have 2 aBM kernels, and we do not have other archaeological samples that could be used to do comparison.

For “It appears that the two kernels were ground separately and that DNA was isolated separately, but data were ultimately pooled across these genetically distinct individuals for analysis”, as you can see from our Materials and Methods section that “Whole kernels were crushed in a mortar and pestle”, these two kernels were ground together before sequenced.

While morphological assessment of the sample would be interesting, most morphological data reported for maize are from microremains (starch, phytoliths, pollen) and this is beyond the scope of our study. Most studies of macrobotanical remains do not appear to focus solely on individual kernels, but instead on (or in combination with) cob and ear shape, which were not available in the assemblage.

(2) Genetic comparison to other samples:

The authors did not meaningfully address the varying ages of the other archaeological samples and modern maize when comparing the genetic distance of their samples. The archaeological samples were as old as >5000 BP to as young as 70 BP and therefore have experienced varying extents of genetic drift from ancestral allele frequencies. For this reason, age should explicitly be included in their analysis of genetic relatedness.

We have changed related part in our revised manuscript.

(3) Assessment of selection in their ancient Bolivian sample:

This analysis relied on the identification of alleles that were unique to the ancient sample and inferred selection based on a large number of unique SNPs in two genes related to internode length. This could be a technical artifact due to poor alignment of sequence data, evidence supporting pseudogenization, or within an expected range of genetic differentiation based on population structure and the age of the samples. More rigor is needed to indicate that these genetic patterns are consistent with selection. This analysis may also be affected by the pooling of the Bolivian archaeological samples.

We do not think it is because of poor alignment of sequence data since we used BWA v0.7.17 with disabled seed (-l 1024) and 0 mismatch alignment. Therefore, there are no SNPs that could come from poor alignment. Please see our detailed methods description here “For the

archaeological maize samples, adapters were removed and paired reads were merged using AdapterRemoval60 with parameters --minquality 20 --minlength 30. All 5' thymine and 3' adenine residues within 5nt of the two ends were hard-masked, where deamination was most concentrated. Reads were then mapped to soft-masked B73 v5 reference genome using BWA v0.7.17 with disabled seed (-l 1024 -o 0 -E 3) and a quality control threshold (-q 20) based on the recommended parameter61 to improve ancient DNA mapping".

For "More rigor is needed to indicate that these genetic patterns are consistent with selection", Could you please be more specific about which method or approach we should use here? For example, methods from specific publications that could be referenced? Or which specific tool could be used?

"This analysis may also be affected by the pooling of the Bolivian archaeological samples." As we could not prove these two seeds came from two different individual plants, we do not think this analysis was affected by the pooling of the Bolivian archaeological samples.

(4) Evidence of selection in modern vs. ancient maize: In this analysis, samples were pooled into modern and ancient samples and compared using the XP-EHH statistic. One gene related to ovule development was identified as being targeted by selection, likely during modern improvement. Once again, ancient samples span many millennia and both South, Central, and North America. These, and the modern samples included, do not represent meaningfully cohesive populations, likely explaining the extremely small number of loci differentiating the groups. This analysis is also complicated by the pooling of the Bolivian archaeological samples.

Yes, it is possible that ovule development might be a modern improvement. We re-wrote this part in our revised manuscript.

Reviewer #1 (Recommendations for the authors):

My suggestion is to address the comments that outline why the methods used or results obtained are not sufficient to support your conclusions. Overall, I suggest limiting the narrative of Inca influence and framing it as speculation in the discussion section. Presenting conclusions of Inca influence in the title and abstract is not appropriate, given the very questionable evidence.

We agree and have changed the title to "Fifteenth century CE Bolivian maize reveals genetic affinities with ancient Peruvian maize".

Reviewer #2 (Recommendations for the authors):

(1) Line 74: Mexicana is another subspecies of teosinte; the distinction is between ssp. mexicana and ssp. parviglumis (Balsas teosinte), not mexicana and teosinte.

We have corrected this in our revised manuscript.

(2) Line 100-102: This is a bit confusing, it cannot have been a symbol of empire "since its first introduction", since its introduction long predates the formation of imperial politics in the region. Reference 17 only treats the late precolonial Inca context, while ref 22 (which cites maize cultivation at 2450 BC, not 3000 BC) makes no reference to ritual/feasting contexts; it simply documents early phytolith evidence for maize cultivation. As such, this statement is not supported by the references offered.

lines 100-102. This point is well taken and was poor prose on our part. We have modified this discussion to reflect both the confusing statement and we have corrected our mistake in age

for reference 22. associated prose has been modified accordingly.

We have corrected them as “Indeed, in the Andes, previous research showed that under the Inca empire, maize was fulfilled multiple contextual roles. In some cases, it operated as a sacred crop” and “...since its first introduction to the region around 2500 BC”.

(3) Line 161: IntCal is likely not the appropriate calibration curve for this region; dates should probably be calibrated using SHCal.

We greatly appreciate this important (and correct) observation. We have completely recalibrated the maize AMS result based on the southern hemisphere calibration curve, discussed the new calibrations, and have also invoked two other AMS dates also subjected to the southern hemisphere calibration on associated material for comparison. We are confident in a 15th century AD/CE age for the maize, most likely mid- to late 15th century.

(4) Lines 167-169: The increase of G and A residues shown in Supplementary Figure S1a is just before the 5' end of the read within the reference genome context, and is related to fragmentation bias - a different process from postmortem deamination. Deamination leads to 5' C->T and 3' G->A, resulting in increased T at 5' ends and increased A at 3' ends, and the diagnostic damage curve. The reduction of C/T just before reads begin is not a result of deamination.

We have removed the “Both features are indicative of postmortem deamination patterns” in our revised manuscript.

(5) Lines 187-196 This section presents a lot of important external information establishing hypotheses, and needs some references.

We have added the related references here.

(6) Line 421: This makes it sound like damage masking was done BEFORE read mapping. However, this conflicts with the previous paragraph about map Damage, and Supplementary Figure 1 still shows a slight but perceptible damage curve, which is impossible if all terminal Ts and As are hard-masked. This should be reconciled.

The Supplementary Figure 1 shows the raw ancient maize DNA sample before damage masking. Specifically, Step1: We used map Damage to check/estimate if the damage exists, and we made the Supplementary Figure 1. Step 2: Then we used our own code hard-masked the damage bases and did read mapping.

The purpose of Supplementary Figure 1 is to show the authenticity of aBM as archaeological maize. Therefore, it should show a slight but perceptible damage curve.

(7) Line 460: PCA method is not given (just the LD pruning and the plotting).

The merged dataset of SNPs for archaeological and modern maize was used for PCA analysis by using “plink -pca”.

(8) "tropicalis" maize is not common usage, it is not clear to me what this refers to.

We have changed all “tropicalis maize” as “tropical maize” in our revised manuscript.

(9) The Figure 4 color palette is not accessible for colorblind/color-deficient vision.

We have changed the color of Figure 4. Please find the new colors in our upload Figure 4.

| (10) *Datasets S2 and S3 are not included with this submission.*

Thank you for letting us know and your suggestion. We have included Datasets S2 and S3 here.

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